CASE FILE • RAPID RESPONSE DETECTIVE SQUAD

THE CASE OF THE SILENT MINERAL

Patient: Raymond "Ray" Kessler, 74 • Room 5E-14, Telemetry Step-Down • Tuesday 0700 to Wednesday 0700

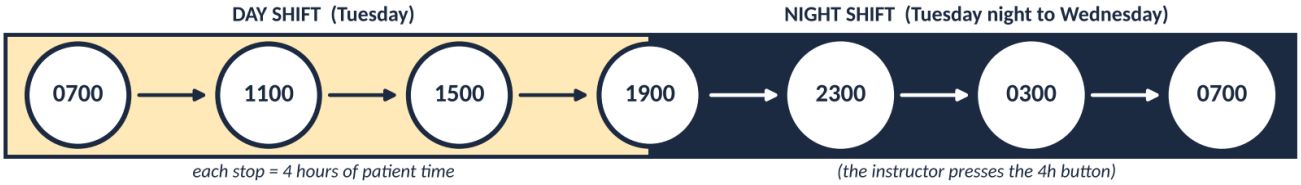
YOUR SQUAD

Squad name (make it good): _____

Name (print)	Starting job at 0700 (A, B or C)	Initials

MISSION BRIEFING

1. **You are the nursing team for Ray Kessler** for 24 hours of patient time: the day shift and then the night shift.
2. **Three jobs, and you rotate them every checkpoint:** (A) Chart Detective, (B) Lab Interpreter, (C) Bedside Nurse. Everybody does every job. Details are on the role card page.
3. **At each checkpoint** the instructor sets the bedside monitor and reads the Situation. You work your role boxes, then hold a **Team Huddle**: top 3 priorities, one question for the provider, one SBAR call.
4. **Do not turn the page until the instructor advances the clock** (the 4h button). Later pages give away later events, and spoilers ruin a good mystery.
5. **Use the EHR on your own device.** Your patient and patient time are synced with the instructor. Small "Where to look" hints tell you where to click.



each stop = 4 hours of patient time *(the instructor presses the 4h button)*

Opening	Six checkpoints	Case Closed huddle	Total
about 10 minutes roles, handoff, patient card	about 15 minutes each 0700, 1100, 1500, 1900, 2300, 0300	about 15 minutes final check, teach-back, debrief	about 2 hours

20 POINTS, MADE SIMPLE

5	6	5	4
Teamwork	Clinical reasoning	Safety, priorities, SBAR	Completeness and reflection

Complete and thoughtful = full credit. Partial credit is generous. The full rubric is on the next page.

TURN-IN

Put the pages in order and fasten them with a **PAPERCLIP (no staples)**. One packet per squad. The reflection strips and the peer check at the end are individual: write your own name on yours.

HOW YOU EARN YOUR 20 POINTS

This is your extra quiz grade. It rewards a great, careful, team-minded effort. There are no trick questions.

4 categories | 3 levels each

Category and points	FULL credit	GOOD (partial)	GETTING THERE (partial)
Teamwork and participation **5 points**	**5 points.** All names on the cover. Roles rotated every checkpoint and initialed. Every teammate clearly contributed in the huddles. Every teammate finished the peer check.	**3 to 4 points.** Roles rotated most of the time (a few initials missing). Most teammates contributed. Peer check done by most.	**1 to 2 points.** One person did most of the work, or roles were not rotated. Peer check missing. (0 = nothing handed in.)
Clinical reasoning at each checkpoint **6 points**	**6 points.** All six checkpoints have thoughtful answers in the Detective, Interpreter and Bedside boxes, using real chart numbers and giving reasons.	**4 to 5 points.** Most checkpoints are thoughtful. One or two are thin (answers without numbers or reasons).	**1 to 3 points.** Many boxes are blank or one-word answers, or the chart facts are mostly missing.
Safety, priorities and SBAR communication **5 points**	**5 points.** At each of the five calls (1100, 1500, 1900, 2300, 0300): safe priority order, a specific question, and a complete SBAR with numbers and a clear request.	**3 to 4 points.** Mostly safe priorities and SBARs. A part is missing sometimes (no numbers, no request).	**1 to 2 points.** Priorities are unsafe or very vague, or SBARs are mostly empty.
Completeness and reflection **4 points**	**4 points.** Every checkpoint page is finished (Mg-o-meter marked, vitals logged, prediction written). Case Closed timeline done. Teach-back to Eleanor done. Individual reflection done.	**2 to 3 points.** One or two of these four parts are partly done.	**0 to 1 point.** Most of the end pages or checkpoint pages are blank.

WHERE EACH POINT COMES FROM

Category	How the points are counted
Teamwork and participation (5)	Roles rotated and initialed = 2. Huddles show the whole team = 2. Peer check done (individual) = 1.
Clinical reasoning at each checkpoint (6)	1 point for each checkpoint (0700, 1100, 1500, 1900, 2300, 0300). Half a point is fine when a checkpoint is mostly done.
Safety, priorities and SBAR communication (5)	1 point for each checkpoint from 1100 to 0300. The 0700 huddle is a warm-up: it is practice, not scored here.
Completeness and reflection (4)	1 point each: pages finished = 1 (group), timeline = 1 (group), teach-back = 1 (group), individual reflection = 1 (individual).

ON EVERY CHECKPOINT PAGE

- A small **points tracker** at the bottom of each checkpoint shows what that page is worth. Each checkpoint is about **1 point of Clinical reasoning** and (from 1100 on) **1 point of Safety, priorities and SBAR**.
- Stretch boxes are marked with a star (★). If time is short, skip the stars first; they are never needed for full credit.
- We look for reasons and chart numbers, not perfect wording. A thoughtful answer that differs from ours still earns credit.

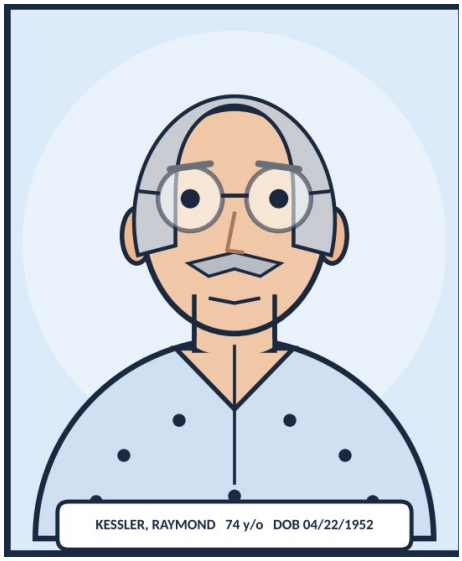


BONUS

answer all six "Predict the plot twist" boxes. There are no wrong answers: a thoughtful guess counts. (Your total cannot go above 20 unless your instructor says otherwise.)

PATIENT CARD: RAYMOND "RAY" KESSLER

Room 5E-14 | Telemetry Step-Down



Name: Raymond "Ray" Kessler, he/him

Age / DOB: 74 years old | DOB 04/22/1952

MRN: SIM-40318

Height / weight: 175 cm | 88.0 kg on arrival (dry weight about 90 kg)

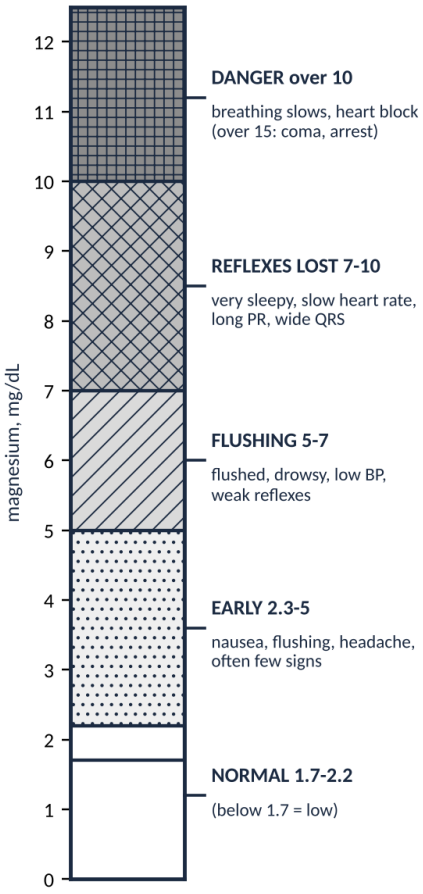
ALLERGY: **Penicillin** (childhood rash)

Code status: **Full Code** | Fall risk: HIGH

Family: Wife **Eleanor Kessler** (72, health care proxy, lives with him); daughter Diane Ruiz (15 minutes away)

Life: Retired machinist. Former smoker (quit 2005). No alcohol.

Care team: Hospitalist Dr. D. Okafor (0700-1900); night hospitalist Dr. J. Lindgren; nephrology Dr. P. Raman; ICU outreach Dr. H. Moreno; pharmacist L. Ferreira



MG-O-METER GUIDE

Normal magnesium is **1.7 to 2.2 mg/dL**. In this lab, anything **over 5.0** is a critical value.

Magnesium leaves the body through the **kidneys**. When the kidneys are failing, magnesium piles up.

High magnesium blocks the signals between **nerves and muscles** and slows the **heart**. The bands on the gauge show what you may see. They overlap a bit in real life: **trust your exam as much as the number.**

HISTORY (from the ED and the clinic)

- **Kidneys:** CKD stage 4. Baseline creatinine 2.6, eGFR 25 (Oct 20). Not yet seen by nephrology.
- **Diabetes** type 2 for 15 years, with neuropathy (A1c 7.8%). **Heart:** HFpEF (EF 55%), high blood pressure, high cholesterol.
- **Other:** heartburn (GERD), chronic low back pain, constipation from his pain medicine.

HOME MEDICINES ON THE ED LIST

- Lisinopril 20 mg daily | Metoprolol succinate 50 mg daily
- Furosemide 20 mg daily | Aspirin 81 mg | Pantoprazole 40 mg
- Insulin glargine 22 units at bedtime (last dose Monday 2100) + lispro 6 units with meals
- Atorvastatin 40 mg at bedtime | Gabapentin 300 mg at bedtime
- Oxycodone 5 mg every 12 hours as needed

(Stores sell more than doctors prescribe. Is the ED list the whole story? Detectives, keep that in mind.)

YOUR CASE FILE STARTS HERE

Ray is 74, he has failing kidneys, and something made him very sleepy and slow. Your squad has 24 hours of patient time to figure out what, stop it, and keep Ray safe. Good detectives write things down.

WHO DOES WHAT: SQUAD ROLE CARDS

rotate every checkpoint

A CHART DETECTIVE

"The chart is the crime scene. Every order and every medicine is a clue."

- Read the **MAR, Orders, notes and sticky notes** for the current time.
- What is **active**, what is **due**, what is **held**?
- What is **dangerous** for Ray? What is **missing**? What should be **stopped**?
- Hunt for **hidden magnesium**: protocols and order sets can hide it in the last step.

WHERE TO CLICK
Orders, MAR, Chart Review (notes), Patient Summary (sticky notes)

B LAB INTERPRETER

"Numbers tell a story if you line them up."

- Compare **new results** to the diagnosis and to the **trend**: magnesium, potassium, calcium, creatinine, blood gas, ECG.
- Say what is **better, worse or the same**, and **why**.
- Plot the new magnesium on the **Mg-o-meter** and name its zone.

WHERE TO CLICK
Lab Results (newest column on the right), ECG interpretation

C BEDSIDE NURSE

"I am the eyes, ears and hands at the bedside."

- Read the **bedside monitor** the instructor sets and **chart the vitals** in the EHR.
- Do and report the **assessment**: level of consciousness, breathing, **reflexes**, strength, urine output.
- Give a **20-second report** to the team: the 3 findings that worry you most.

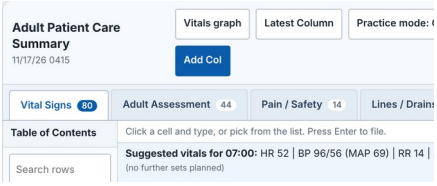
WHERE TO CLICK
Flowsheets (Vital Signs, Adult Assessment, Intake/Output), the monitor

ROTATION CHART (write your names, then follow the letters)

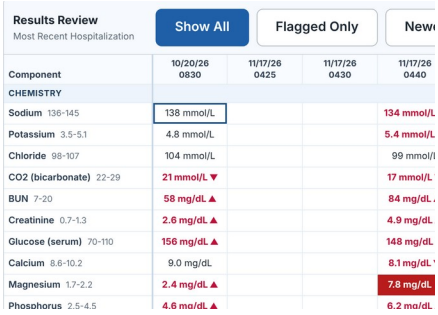
Checkpoint	Student 1: _____	Student 2: _____	Student 3: _____
0700 (1)	A	B	C
1100 (2)	B	C	A
1500 (3)	C	A	B
1900 (4)	A	B	C
2300 (5)	B	C	A
0300 (6)	C	A	B

Squad of two? One of you is the **Desk** (does A and B) and the other is the **Bedside Nurse** (C). Swap jobs at every checkpoint (Student 1 is the Desk at 0700, 1500 and 2300; Student 2 at 1100, 1900 and 0300). Skip the starred (★) stretch boxes if time is short.

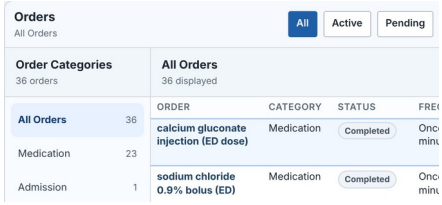
WHERE TO CLICK IN THE EHR (tiny hints; click the tab named in each box)



C: Flowsheets > Vital Signs > "Add Col". Type or pick your monitor values.



B: Lab Results. Newest column is on the right. Red = abnormal.



A: Orders (click one to read its details); MAR has a PRN filter and "due" colors.

Turn in with a paperclip (no staples). Write in dark pen. Do not turn the page until the instructor advances the clock.

OPENING: HANDOFF FROM THE ED

Tuesday 0655

The instructor reads the ED nurse's report aloud (K. Ortiz, RN). Listen, write, and then read it back. You are the day shift taking over Ray.

S B A R listening notes (jot short phrases)	Write here
S Situation Who is this and what is wrong right now?	
B Background How did he get here? What did the ED do, and when?	
A Assessment (0700) What did the ED nurse see at 0700?	
R Recommendation: what is due next? What is still to do? What worries you?	

ED FACT CARD (printed for you; the ED nurse will not repeat every number)

ED vitals 0425: T 97.2, HR 46, BP 82/46, RR 14, SpO2 95% room air, glucose 148.

ED labs (drawn 0440): Mg **7.8**, K 5.4, creatinine 4.9 (baseline 2.6), BUN 84, eGFR 12, CO2 17, Ca 8.1, phosphorus 6.2.

ED ECG 0430: sinus bradycardia 46, **PR 248 ms**, **QRS 124 ms**, QTc 468.

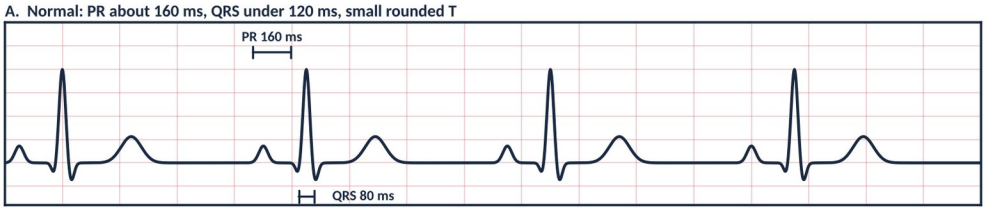
ED treatments: ondansetron 4 mg IV 0510; calcium gluconate 1 g IV 0555-0605; NS 500 mL 0605-0635; Foley 0610.

☐ **Read-back done:** name, date of birth and allergy checked against the wristband

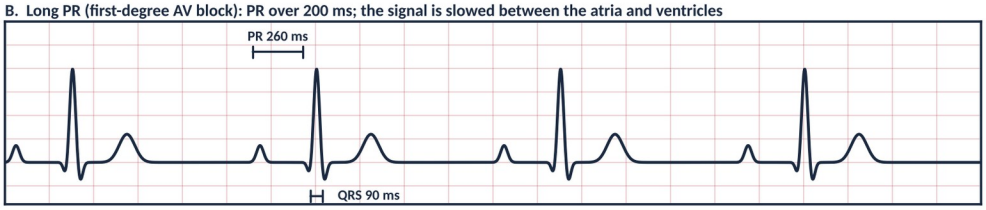
☐ **Asked one clarifying question** of the ED nurse (write it): _____

RHYTHM DECODER (keep this page handy all shift: strips A, B and C show what magnesium does to the ECG)

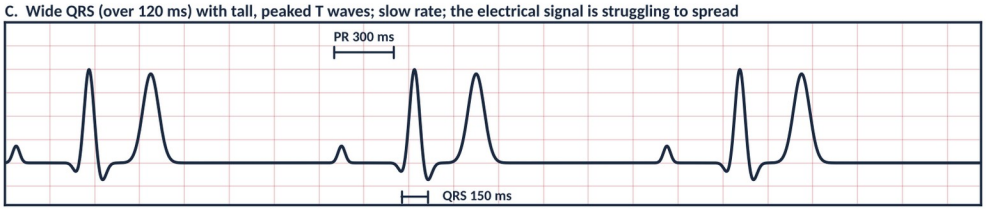
A. Normal: PR about 160 ms, QRS under 120 ms, small rounded T



B. Long PR (first-degree AV block): PR over 200 ms; the signal is slowed between the atria and ventricles



C. Wide QRS (over 120 ms) with tall, peaked T waves; slow rate; the electrical signal is struggling to spread



Teaching examples (not Ray's own strips). Each large red box = 0.2 second.

MISSION 1: First Look

Something is slowing Ray down. Find it, name it, and stop feeding it.

Tuesday 0700 | Opening scene

SITUATION (the instructor reads this aloud)

It is **Tuesday 0700** on 5 East, room 5E-14. Ray Kessler, 74, came up from the ED about ten minutes ago. He has kidney disease, diabetes and a weak heart. He vomited for four days and then could not be woken this morning.

The ED found a **very high magnesium (7.8; normal is 1.7 to 2.2)** and a creatinine of 4.9. They gave calcium gluconate, a saline bolus and placed a Foley. His wife Eleanor sits by the bed holding a paper bag from home. She has not been asked about it yet.

C BEDSIDE NURSE

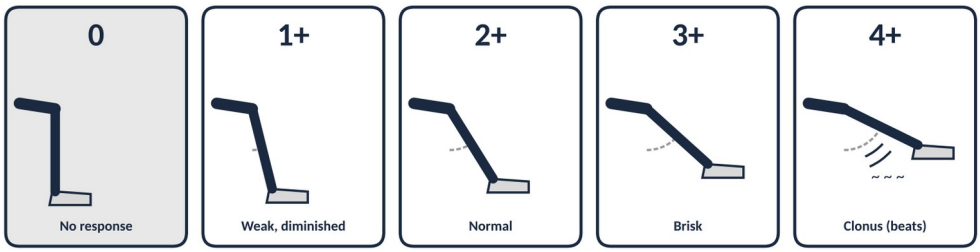
Read the monitor, chart it, examine Ray. Then give your team a 20-second report.

Initials: _____

Monitor vitals (the instructor sets the bedside monitor to 0700; read your own set and chart it in the EHR).

Time	Temp (F)	HR	BP (MAP)	RR	SpO2 / oxygen	Pain	Rhythm (read the monitor)	POC glucose
last time ED 0425	97.2	46	82/46 (58)	14	95% RA	-	Sinus bradycardia (ECG 0430)	148
0700 NOW								
trend ↑ ↓ =								

What to check	What you find (write it)	Alarm?
Level of consciousness (RASS) and orientation		☐
Breathing: rate for a FULL minute, depth, pattern		☐
Reflexes: patellar and biceps (0 to 4+)		☐
Strength: grip and lifting the legs		☐
Lungs (before any fluid!) and neck veins		☐
Skin and mouth		☐
Urine (Foley): color and amount		☐



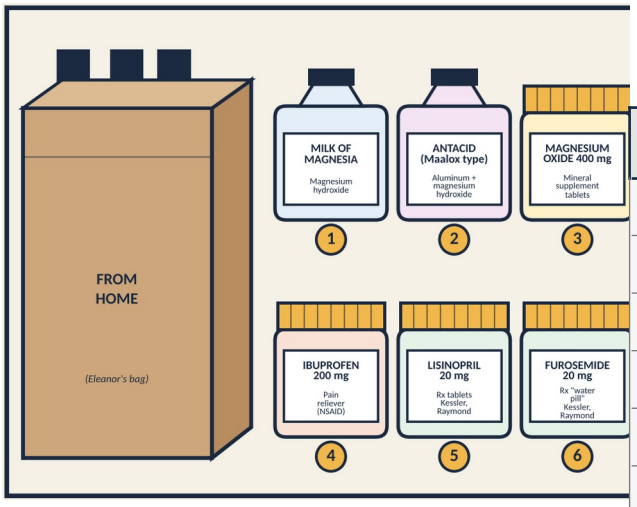
Reflex scale: tap below the kneecap. 0 none, 1+ weak, 2+ normal, 3+ brisk, 4+ clonus.

1C1 Your 20-second report. Say it out loud to the team, then write it: the **three findings** that worry you most, and why.

A CHART DETECTIVE

Read the chart like a crime scene: orders, MAR, notes, sticky notes.

Initials: _____



1A1 The evidence bag. Eleanor opens the bag (the instructor tells you when: ask her about over-the-counter products!). For each bottle, read the label and fill in the table.

#	Active ingredient (from the label)	Mg? Y/N	Why it matters for Ray
1			
2			
3			
4			
5			
6			

1A2 MAR and orders sweep. **(a)** The saline bolus is due at 0715. What must your team do before it starts and after it ends? (Ray has a weak heart: HFpEF.) **(b)** DETECTIVE HUNT: read the Active orders all the way down, including standing "order set" protocols. Is anything there with **magnesium, phosphate or a sedative**? Write what you found and where, or "not found yet". *Where to look: MAR (Due) for (a); Orders > Active, click an order to open its details, for (b)*

B LAB INTERPRETER

Turn the numbers into a story: compare to the diagnosis and the trend.

Initials: _____

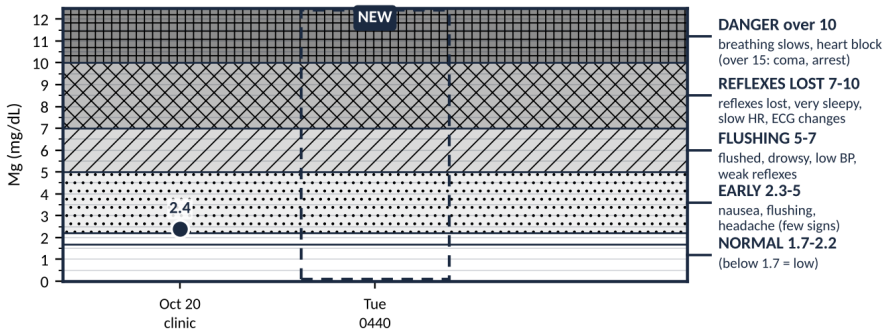
1B1 Copy the labs. Copy the 10/20 clinic and ED 0440 values. Then write **H** (higher than normal) or **L** (lower). *Where to look: Lab Results: newest column is on the right; click Show All*

Test	Normal	Oct 20 clinic	Tue 0440 (ED)	H or L?
Magnesium (mg/dL)	1.7-2.2			
Potassium (mmol/L)	3.5-5.1			
Creatinine (mg/dL)	0.7-1.3			
Calcium (mg/dL)	8.6-10.2			

1B2 Rhythm decoder. The ED ECG (0430): HR 46, **PR 248 ms**, **QRS 124 ms**. At 0700 the monitor shows PR 0.26 s. Which strip on the Handoff page is closest, **A, B or C**? Finish: "Magnesium makes the heart...".

1B3 Connect the dots. In your own words: why can Ray's magnesium climb so high? Use **two numbers** from your table.

MG-O-METER: mark it, name the zone, say what it means



Mark an X at the new level, then write its zone below. Bands overlap a little in real life: trust your exam as much as the number.

1M New level (Tue 0440): _____ mg/dL Zone: _____

What this level means for Ray (use the bands and what you saw at the bedside):

TEAM HUDDLE: everybody talks, one person writes — about 4 minutes

1T1 Top 3 priorities. List the **top 3 priorities for Ray's first hour**, in order, with a few words on why.

#	Priority (in order)	Why (a few words)
1		
2		
3		

1T2 Our ONE question. we will ask the provider (make it specific to Ray):

1P Predict the plot twist (bonus). In the next 4 hours, what do you think will happen to Ray's **breathing and magnesium level**? Why?

1T3 SBAR call. Practice call: **tell the hospitalist what you found in the bag.** (Use S-B-A-R.)

	Say it out loud, then write it (short phrases and numbers)
S	
B	
A	
R	

CLUE UNLOCKED #1

Kidneys are the exit door for magnesium. When the door is almost shut (eGFR 12), every swallow of a magnesium product adds to the pile. Write it on your case file: **Suspect = the bag.**

POINTS ON THIS PAGE

- Clinical reasoning: ___ / 1 (of 6)
- Safety / SBAR: warm-up, not scored
- Teamwork, Complete: whole packet

STOP

Wait for the instructor's 4h button

MISSION 2: Code Yellow

Tuesday 1100 | 0700 to 1100

The clues are getting louder. Read them quickly and call for help early.

SITUATION (the instructor reads this aloud)

It is **1100**. Since 0700 Ray has had another saline bolus, maintenance saline, and a water pill (furosemide) by IV at 0915. The pharmacist went through Eleanor's bag and the home products were stopped.

The 0900 labs are back: **magnesium 8.7**, up from 7.8. Eleanor stepped out to call their daughter. You walk in to check on Ray.

C BEDSIDE NURSE

Read the monitor, chart it, examine Ray. Then give your team a 20-second report.

Initials: _____

Monitor vitals (the instructor sets the bedside monitor to 1100; read your own set and chart it in the EHR).

Time	Temp (F)	HR	BP (MAP)	RR	SpO2 / oxygen	Pain	Rhythm (read the monitor)	POC glucose
last time 0700	97.6	52	96/56 (69)	14	94% RA	2/10	Sinus bradycardia with first-degree AV block (PR 0.26 s)	132
1100 NOW								
trend ↑ ↓ =								

What to check	What you find (write it)	Alarm?
Level of consciousness (RASS) and orientation		☐
Breathing: rate for a FULL minute, depth, pattern		☐
Reflexes: patellar and biceps (0 to 4+)		☐
Strength: grip and lifting the legs		☐
Lungs and neck veins		☐
Skin, mouth and pupils		☐
Urine (Foley): color and amount		☐

2C1 Call for help? Look at your numbers and your exam. Is this a "call for help NOW" moment? Name the **numbers or findings** that tell you.

2C2 The oxygen trap. You put Ray on oxygen and the SpO2 climbs. Did that fix the problem? Why or why not?

A CHART DETECTIVE

Read the chart like a crime scene: orders, MAR, notes, sticky notes.

Initials: _____

2A1 Call numbers. Find the nursing order for vital signs, level of consciousness, RR and reflexes. Copy its **"call the provider for..."** list. How many of those are true for Ray right now? *Where to look: Orders > Nursing (or filter Active)*

2A2 The antidote order. Find the PRN **calcium gluconate** order. Write: dose, how fast, and what the nurse must check before and during. *Where to look: Orders or MAR > PRN*

2A3 What's missing? A deteriorating patient needs more than Ray has in the chart right now. List **3 things that are NOT yet ordered or set up** and could save his life. *Where to look: Orders (Active) and Devices on Patient Summary*

B LAB INTERPRETER

Turn the numbers into a story: compare to the diagnosis and the trend.

Initials: _____

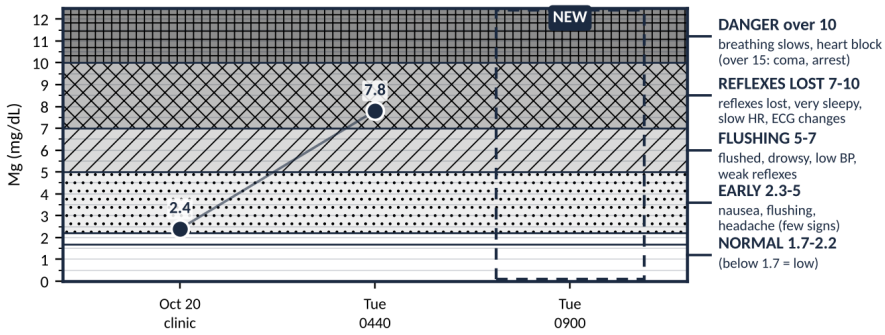
2B1 Update the labs. Copy the ED 0440 values (from your last page) and the new 0900 values. Draw an arrow: up, down or same. *Where to look: Lab Results (newest on the right)*

Test	Normal	Tue 0440	Tue 0900	Arrow
Magnesium (mg/dL)	1.7-2.2			
Potassium (mmol/L)	3.5-5.1			
Creatinine (mg/dL)	0.7-1.3			
Calcium (mg/dL)	8.6-10.2			

2B2 Why is it still rising? Ray got about a liter of saline and a big dose of furosemide, yet Mg went **up** (7.8 to 8.7). Also read the new urine test: **FENa 0.6%** (under 1% = "prerenal"). What does all this say about why the magnesium is not leaving? *Where to look: Lab Results > Urine / Chemistry (FENa); Chart Review > Renal ultrasound note*

2B3 ★ Rhythm check. The monitor now shows PR **0.30 s** (0700: 0.26; ED ECG: 248 ms). Longer or shorter? Which strip on the Handoff page is it heading toward? What do you want the team to have ready?

MG-O-METER: mark it, name the zone, say what it means



Mark an X at the new level, then write its zone below. Bands overlap a little in real life: trust your exam as much as the number.

2M New level (Tue 0900): _____ mg/dL Zone: _____

What this level means for Ray (use the bands and what you saw at the bedside):

TEAM HUDDLE: everybody talks, one person writes — about 4 minutes

2T1 Top 3 priorities. Top 3 priorities right now, in order. (Think airway, breathing, circulation.)

#	Priority (in order)	Why (a few words)
1		
2		
3		

2T2 Our ONE question. for the rapid response doctor or hospitalist:

2P Predict the plot twist (bonus). you ask for calcium. It works for about half an hour to an hour. What do you think the NEXT magnesium level will show, and what will the team need to do?

2T3 SBAR call. SBAR call to the rapid response team / hospitalist.

	Say it out loud, then write it (short phrases and numbers)
S	
B	
A	
R	

CLUE UNLOCKED #2

Calcium **protects**, it does not **remove**. It buys 30 to 60 minutes while magnesium is held back from the nerves and heart. Something has to take the magnesium OUT of the body. What could that be?

POINTS ON THIS PAGE

- Clinical reasoning: ___ / 1 (of 6)
- Safety, priorities, SBAR: ___ / 1 (of 5)
- Teamwork, Complete: whole packet

STOP

Wait for the instructor's 4h button

MISSION 3: The Big Decision

When medicine alone is not enough. Check every box before the machine starts.

Tuesday 1500 | 1100 to 1500

SITUATION (the instructor reads this aloud)

It is **1500**. The rapid response team came at 1115. Ray got calcium gluconate (1 g at 1120, and 2 g more ordered at 1235 and again at 1400). Nephrology ordered a big dose of furosemide, but he made only 70 mL of urine in four hours.

At 1400 the lab called: **magnesium 10.9**, potassium 5.8, creatinine 5.3. The 1330 ECG showed HR 38 and a wider QRS. Dr. Raman decided on **emergency dialysis**. Eleanor gave phone consent at 1350 with two nurses listening. The dialysis catheter went into the right side of his neck at 1440 and the x-ray is clear. **Dialysis starts now.** ICU outreach says he can stay on 5 East with a 1:1 nurse.

C BEDSIDE NURSE

Read the monitor, chart it, examine Ray. Then give your team a 20-second report.

Initials: _____

Monitor vitals (the instructor sets the bedside monitor to 1500; read your own set and chart it in the EHR).

Time	Temp (F)	HR	BP (MAP)	RR	SpO2 / oxygen	EtCO2	Pain	Rhythm (read the monitor)	POC glucose
last time 1100	97.4	44	84/48 (60)	10, shallow	91% RA	-	0/10	Sinus bradycardia, long PR (0.30 s)	104
1500 NOW									
trend ↑ ↓ =									

What to check	What you find (write it)	Alarm?
Level of consciousness (RASS) and orientation		☐
Breathing: rate, depth, EtCO2		☐
Reflexes: patellar and biceps (0 to 4+)		☐
Strength: grip and lifting the legs		☐
Lungs and neck veins		☐
Neck catheter site		☐
Urine (Foley): color and amount		☐

3C1 Breathing watch. Write your RR, SpO2 and EtCO2 trend (0700, 1100, now). If Ray's RR drops to 7 during dialysis, what do you do **in the first minute**?

3C2 New crackles. Crackles at both bases are NEW since 0700. What do you suspect? What do you **not** want to give, and who do you tell?

A CHART DETECTIVE

Read the chart like a crime scene: orders, MAR, notes, sticky notes.

Initials: _____

3A1 Dialysis readiness check. For each item write **where you found it** in the chart, or mark it **missing**. *Where to look: Chart Review (notes), Orders, Devices on Patient Summary, Lab Results*

Check before the machine starts	Found? Where?
Consent: who gave it, how, who witnessed?	
The dialysis order (how long, what kind of bath, heparin?)	
Catheter placed and chest x-ray read BEFORE first use	
Calcium gluconate 2 g before dialysis (time due)	
Maintenance saline stopped (time)	
Blood-pressure medicines held	

3A2 Catheter rules. The new neck catheter is for dialysis. Write **two things you may do** and **two things you must never do** with it. *Where to look: Orders > Nursing: "Dialysis catheter: use by dialysis staff only"*

3A3 ICU thresholds. Find the ICU outreach note (written 1435). Copy **three numbers or findings** that mean "intubate and transfer to the ICU". *Where to look: Chart Review > Notes*

B LAB INTERPRETER

Turn the numbers into a story: compare to the diagnosis and the trend.

Initials: _____

3B1 Trend the labs. Copy the three Mg draws and the other numbers. Then circle the one that worries you most. *Where to look: Lab Results (Tue 0900, 1115, 1315 columns); ABG under Blood Gas*

Test	Normal	Tue 0900	Tue 1115	Tue 1315
Magnesium (mg/dL)	1.7-2.2			
Potassium (mmol/L)	3.5-5.1			
Creatinine (mg/dL)	0.7-1.3			
CO2 (mmol/L)	22-29			

3B2 Blood gas and breathing. Copy the two blood gases. **What do they tell you about Ray's breathing, and why is oxygen not the fix?** *Where to look: Lab Results > Blood Gas (ABG 1130 and 1415)*

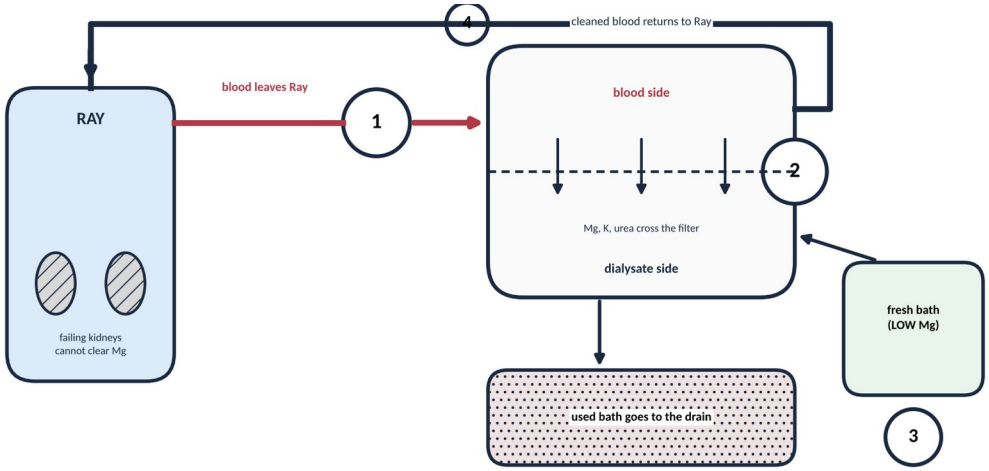
ABG	Normal	1130 (on 3 L)	1415 (on 4 L)
pH	7.35-7.45		
PaCO2 (mmHg)	35-45		
PaO2 (mmHg)	80-100		
HCO3 (mmol/L)	22-26		

B LAB INTERPRETER

Continued: how the machine works.

Initials: _____

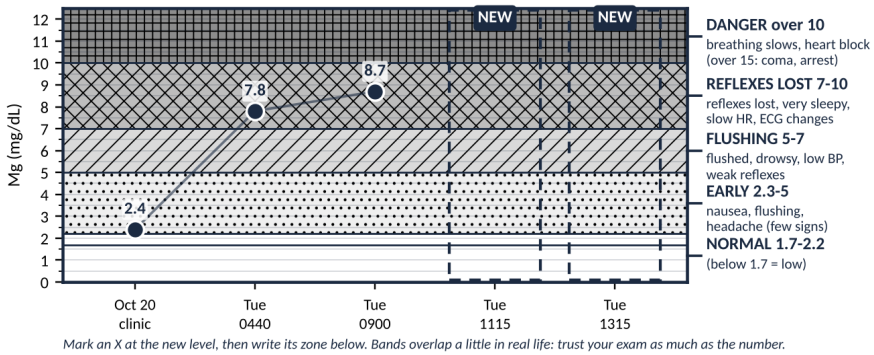
3B3 Heart strip. ECG at 1330: HR 38, **PR 320 ms, QRS 148 ms, tall peaked T waves.** Which strip (A, B or C)? Name **two** reasons T waves can be tall in Ray (hint: two labs).



3B4 ★ Label the machine. Name parts 1 to 4 from the word bank, then finish: "Magnesium leaves the blood because…". Word bank: blood pump / dialyzer filter (membrane) / cleaned blood returns / fresh bath low in magnesium

#	What it is
1	
2	
3	
4	
Why?	

MG-O-METER: mark it, name the zone, say what it means



Mark an X at the new level, then write its zone below. Bands overlap a little in real life: trust your exam as much as the number.

3M New levels (1115 and 1315): _____ and _____ mg/dL Zone for the higher one: _____

What this level means for Ray (use the bands and what you saw at the bedside):

TEAM HUDDLE: everybody talks, one person writes — about 4 minutes

3T1 Top 3 priorities. **Top 3 priorities** while dialysis starts, in order.

#	Priority (in order)	Why (a few words)
1		
2		
3		

3T2 Our ONE question. for the ICU outreach team / nephrology:

3P Predict the plot twist (bonus). dialysis runs for about 3 hours. What will the magnesium be afterward, and what **new problems** might a 3-hour dialysis session cause?

3T3 SBAR call. **SBAR call to ICU outreach / nephrology** (his breathing is slowing before dialysis even starts).

	Say it out loud, then write it (short phrases and numbers)
S	
B	
A	
R	

CLUE UNLOCKED #3

Dialysis is the exit door that works when the kidneys cannot. It can drop magnesium fast, **but fast changes can cause their own problems.** Keep your detective eyes open for what a 3-hour session might do to the body.

POINTS ON THIS PAGE

- Clinical reasoning: ___ / 1 (of 6)
- Safety, priorities, SBAR: ___ / 1 (of 5)
- Teamwork, Complete: whole packet



Wait for the instructor's 4h button

MISSION 4: The Quiet After the Storm

Ray looks so much better. Great detectives keep looking.

Tuesday 1900 | 1500 to 1900

SITUATION (the instructor reads this aloud)

It is **1900**. Dialysis ran from 1500 to 1800 and took off 500 mL of fluid. At 1630 his blood pressure dipped to 82/46; the dialysis nurse gave 250 mL of saline and slowed the machine. By 1700 his magnesium was 7.2 and after dialysis (1800) it was **4.6**. Ray is awake and breathing 16 times a minute. The 1850 results are in. **Your squad rolls straight into the night shift** as the day team gives report. Look him over.

C BEDSIDE NURSE

Read the monitor, chart it, examine Ray. Then give your team a 20-second report.

Initials: _____

Monitor vitals (the instructor sets the bedside monitor to 1900; read your own set and chart it in the EHR).

Time	Temp (F)	HR	BP (MAP)	RR	SpO2 / oxygen	Pain	Rhythm (read the monitor)	POC glucose
last time 1500	97.2	46	88/50 (63)	9	92% on 4 L nasal cannula	CPOT	Sinus bradycardia, long PR (0.32 s), wide QRS (0.14 s), tall T waves	88
1900 NOW								
trend ↑ ↓ =								

What to check	What you find (write it)	Alarm?
Level of consciousness and orientation		☐
Breathing: rate for a FULL minute, depth, pattern		☐
Reflexes: patellar and biceps (0 to 4+)		☐
Lungs and swallow screen		☐
Mouth, hands and how he feels		☐
Neck catheter site		☐
Urine (Foley): color and amount		☐

4C1 Post-dialysis watch. The dialysis RN says: "Check Ray every 15 minutes for the next hour." Name **four things** you will check or protect.

4C2 If glucose is low. When you check his glucose at 1905 the instructor tells you the number. **Write the number, the steps you take, and what you chart.**

A CHART DETECTIVE

Read the chart like a crime scene: orders, MAR, notes, sticky notes.

Initials: _____

4A1 The glucose trend. Copy the **serum glucose** at each draw. Then answer: was any insulin given? What did the numbers predict? *Where to look: Lab Results > Glucose (serum); MAR > insulin*

Tue 0440	Tue 0900	Tue 1115	Tue 1315	Tue 1700

4A2 Insulin orders. Which insulin orders are active right now? What do you want the provider to **hold or change**, and why? *Where to look: Orders > Medication; MAR*

4A3 Protocol and juice. Find the nurse-driven **hypoglycemia protocol** order. Write its steps. Then name **one juice that is a poor choice** for Ray, and why. *Where to look: Orders > Medication: "hypoglycemia protocol"*

B LAB INTERPRETER

Turn the numbers into a story: compare to the diagnosis and the trend.

Initials: _____

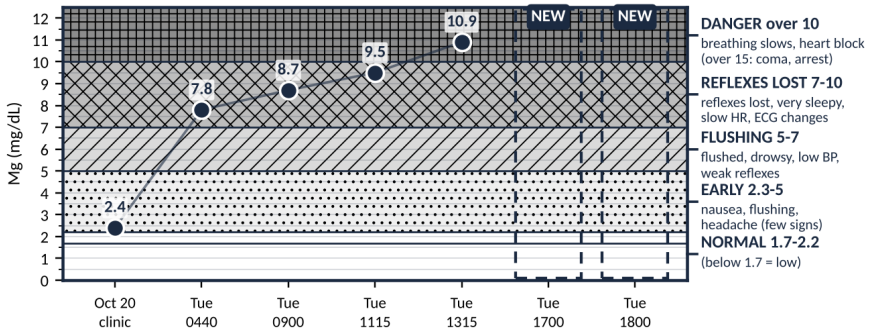
4B1 After the machine. Copy the numbers. **Which ones improved? Which one is now too low?** *Where to look: Lab Results (1315, 1700, 1800 columns); ECG under Cardiac*

Test	Normal	Tue 1315	Tue 1700	Tue 1800
Magnesium (mg/dL)	1.7-2.2			
Potassium (mmol/L)	3.5-5.1			
Creatinine (mg/dL)	0.7-1.3			
Calcium (mg/dL)	8.6-10.2			
Ionized calcium	1.12-1.32			

4B2 Calcium clue. Ray says his mouth and hands tingle. The 1830 ECG shows **QTc 486**. How do those fit together? What would you report, and when would calcium be given?

4B3 ★ Rhythm check. Look at the monitor rhythm (PR 0.20 s) and the 1830 ECG (PR 204, QRS 108). Which strip is it most like now, and what got better?

MG-O-METER: mark it, name the zone, say what it means



Mark an X at the new level, then write its zone below. Bands overlap a little in real life: trust your exam as much as the number.

4M New levels (1700 and 1800): _____ and _____ mg/dL Zone for the lower one: _____

What this level means for Ray (use the bands and what you saw at the bedside):

TEAM HUDDLE: everybody talks, one person writes — about 4 minutes

4T1 Top 3 priorities. Top 3 priorities right now, in order.

#	Priority (in order)	Why (a few words)
1		
2		
3		

4T2 Our ONE question. for the provider:

4P Predict the plot twist (bonus). the magnesium is 4.6 now. What do you think the 2100 level will be, and why might it **not** stay at 4.6?

4T3 SBAR call. SBAR call to the hospitalist about the low glucose.

	Say it out loud, then write it (short phrases and numbers)
S	
B	
A	
R	

CLUE UNLOCKED #4

Magnesium is not only in the blood. A lot of it hides in **cells, bone and the bowel**. When the blood level drops fast, stored magnesium can drift back out. Remember this: **the number can come back.**

POINTS ON THIS PAGE

- Clinical reasoning: ___ / 1 (of 6)
- Safety, priorities, SBAR: ___ / 1 (of 5)
- Teamwork, Complete: whole packet

STOP

Wait for the instructor's 4h button

MISSION 5: The Night Shift Riddle

Tuesday night 2300 | 1900 to 2300

Ray is not himself. Who or what is making him confused, and what will you NOT give?

SITUATION (the instructor reads this aloud)

It is **2300**. D10W (sugar water) has run since 1930 at 40 mL/h and Ray's sugars have been 98, 106 and 118. At 2100 his **magnesium was 5.6**, up from 4.6, and his calcium was 7.9. Eleanor went home at 2100.

Since 2230 Ray has been restless and is pulling at his lines. You go in.

C BEDSIDE NURSE

Initials: _____

Read the monitor, chart it, examine Ray. Then give your team a 20-second report.

Monitor vitals (the instructor sets the bedside monitor to 2300; read your own set and chart it in the EHR).

Time	Temp (F)	HR	BP (MAP)	RR	SpO2 / oxygen	Pain	Rhythm (read the monitor)	POC glucose
last time 1900	97.6	66	104/60 (75)	16	96% RA	3/10	Normal sinus rhythm 66, PR 0.20 s	54 (at 1905)
2300 NOW								
trend ↑ ↓ =								

CAM check for delirium (positive when the first two AND one of the last two are present):

- ☐ Acute change or ups and downs in thinking
- ☐ Disorganized thinking
- ☐ Inattention (cannot keep focus)
- ☐ Altered level of consciousness

Our CAM result: ☐ **Positive** ☐ **Negative**

What to check	What you find (write it)	Alarm?
Level of consciousness; CAM features (see box)		☐
Breathing: rate for a FULL minute, depth, pattern		☐
Reflexes: patellar and biceps (0 to 4+)		☐
Strength / moving in bed		☐
Lungs and neck veins		☐
Lines and tubes (what is he pulling at?)		☐
Urine (Foley): color and amount		☐

5C1 Safe and calm. List **five non-drug things** you will do right now to keep Ray safe and calm, including how you protect his two lines (Foley and neck catheter).

5C2 Do not hide the clues. Even with delirium, which three checks do you **keep doing** (they warn of magnesium trouble)? Why can sleepiness hide them?

Turn in with a paperclip (no staples). Write in dark pen. Do not turn the page until the instructor advances the clock.

A CHART DETECTIVE

Read the chart like a crime scene: orders, MAR, notes, sticky notes.

Initials: _____

5A1 Suspect medicines. Scan the **Orders and MAR** for anything that could cause or worsen confusion, or hide breathing and reflex checks. Name each one and say what you would ask the provider to do. *Where to look: Orders > Medication (look for "as needed for sleep"); MAR > PRN*

5A2 Reversible causes checklist. For each cause, write what the **chart or monitor says** in a few words. *Where to look: Flowsheets, Lab Results, MAR, Chart Review*

Cause of delirium to check	What the chart says
G	l
B	l
B	o
O	x
N	e
L	a

5A3 Bowel plan. PEG and senna were ordered at 2130. Why does constipation matter in a magnesium case, and which two kinds of products must stay away?

B LAB INTERPRETER

Turn the numbers into a story: compare to the diagnosis and the trend.

Initials: _____

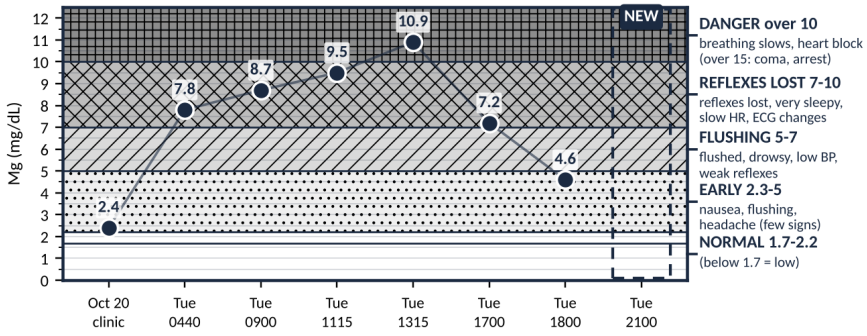
5B1 New labs. Copy the 1800 (after dialysis) and 2100 values. Draw an arrow for the change. *Where to look: Lab Results (Tue 1800 and Tue 2100 columns)*

Test	Normal	Tue 1800	Tue 2100	Arrow
Magnesium (mg/dL)	1.7-2.2			
Potassium (mmol/L)	3.5-5.1			
Creatinine (mg/dL)	0.7-1.3			
Calcium (mg/dL)	8.6-10.2			

5B2 What is going on with Mg? Magnesium went 10.9, then 4.6 (1800), now 5.6. **Explain what you think is happening** in plain words. Where could the extra magnesium be coming from?

5B3 ★ Which numbers could be muddling his brain? Pick **two** labs from your table that could add to confusion, and say how.

MG-O-METER: mark it, name the zone, say what it means



Mark an X at the new level, then write its zone below. Bands overlap a little in real life: trust your exam as much as the number.

5M New level (Tue 2100): _____ mg/dL Zone: _____

What this level means for Ray (use the bands and what you saw at the bedside):

TEAM HUDDLE: everybody talks, one person writes — about 4 minutes

5T1 Top 3 priorities. **Top 3 priorities** right now, in order.

#	Priority (in order)	Why (a few words)
1		
2		
3		

5T2 Our ONE question. for the night hospitalist:

5P Predict the plot twist (bonus). Ray has not had a stool in four days and his magnesium is creeping up. What do you think the next lab will show, and what might Ray ask for around 0300?

5T3 SBAR call. **SBAR call to the night hospitalist about new confusion.**

	Say it out loud, then write it (short phrases and numbers)
S	
B	
A	
R	

CLUE UNLOCKED #5

The riddle has a **hidden accomplice**: a "helpful" order that looks routine. Routine orders and standing protocols are written for ordinary patients. **Always read the ingredients of every order in a kidney patient.**

POINTS ON THIS PAGE

- Clinical reasoning: ___ / 1 (of 6)
- Safety, priorities, SBAR: ___ / 1 (of 5)
- Teamwork, Complete: whole packet

STOP

Wait for the instructor's 4h button

MISSION 6: The Boomerang

The suspect is back, and this time it is asking for help.

Wednesday 0300 | 2300 to 0300

SITUATION (the instructor reads this aloud)

It is **0300 Wednesday**. Ray slept a little. The 0100 magnesium came back **6.4** (4.6 after dialysis, 5.6 at 2100). Calcium is 8.1. The D10W stopped at 0130 and his glucose is 126.

Ray has had no stool in four days and his belly looks full. You go to check him. He mumbles something about "the pink milk".

C BEDSIDE NURSE

Read the monitor, chart it, examine Ray. Then give your team a 20-second report.

Initials: _____

Monitor vitals (the instructor sets the bedside monitor to 0300; read your own set and chart it in the EHR).

Time	Temp (F)	HR	BP (MAP)	RR	SpO2 / oxygen	Pain	Rhythm (read the monitor)	POC glucose
last time 2300	98.2	62	112/64 (80)	16	96% RA	3/10	Sinus rhythm 62, PR 0.22 s	118
0300 NOW								
trend ↑ ↓ =								

What to check	What you find (write it)	Alarm?
Level of consciousness; CAM (see box)		☐
Breathing: rate for a FULL minute, depth, pattern		☐
Reflexes: patellar and biceps (0 to 4+)		☐
Strength: grip and lifting the arms		☐
Lungs and neck veins		☐
Abdomen: size, bowel sounds, tenderness		☐
Urine (Foley): color and amount		☐

6C1 "Can I have the pink milk?" Ray says: "My belly is so full. Can I take something? At home I take the pink milk." **What do you say and do?** (Think: what is the pink milk, and what is in your protocol?)

6C2 Stay ahead of the number. RR 12, HR 56, reflexes 1+. Compare with the call numbers in the vitals order (RR under 12, HR under 45, SBP under 90, reflexes lost, any drop in alertness). **Is he there yet? How often will you recheck, and what makes you call a rapid response?**

A CHART DETECTIVE

Read the chart like a crime scene: orders, MAR, notes, sticky notes.

Initials: _____

6A1 Hidden magnesium hunt. Open the nurse-driven **bowel protocol** order and read it to the very end. Write **each step**, circle the step that is dangerous for Ray, and explain. *Where to look: Orders > Nursing > Bowel protocol (click the order to open its details)*

6A2 Two orders disagree. Find the **active precaution order** about magnesium and phosphate. Both it and the bowel protocol are active. **What do you do about this conflict, and who do you call?** *Where to look: Orders > Precautions*

6A3 ★ A safe bowel plan. List what is **safe** for Ray's bowels and what is **not**. Include one enema. *Where to look: Orders > Medication*

B LAB INTERPRETER

Turn the numbers into a story: compare to the diagnosis and the trend.

Initials: _____

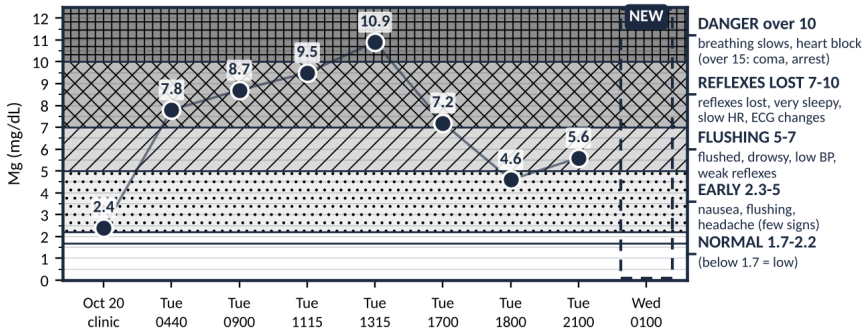
6B1 Three levels in a row. Copy the three most recent magnesium results and the other new numbers. *Where to look: Lab Results (Tue 1800, Tue 2100, Wed 0100 columns)*

Test	Normal	Tue 1800	Tue 2100	Wed 0100
Magnesium (mg/dL)	1.7-2.2			
Potassium (mmol/L)	3.5-5.1			
Creatinine (mg/dL)	0.7-1.3			
Calcium (mg/dL)	8.6-10.2			
Ionized calcium	1.12-1.32			

6B2 What do the last three tell you? Write what the 4.6, 5.6, 6.4 pattern is called and what it means for the **next few hours**.

6B3 Constipation clue. How can constipation keep magnesium high? And the ECG at 0230 shows PR 236 ms (1830 was 204): longer or shorter, and what does that suggest?

MG-O-METER: mark it, name the zone, say what it means



Mark an X at the new level, then write its zone below. Bands overlap a little in real life: trust your exam as much as the number.

6M New level (Wed 0100): _____ mg/dL Zone: _____

What this level means for Ray (use the bands and what you saw at the bedside):

TEAM HUDDLE: everybody talks, one person writes — about 4 minutes

6T1 Top 3 priorities. Top 3 priorities, in order.

#	Priority (in order)	Why (a few words)
1		
2		
3		

6T2 Our ONE question. for the night hospitalist / nephrology:

6P Predict the plot twist (bonus). by 0700 Ray has had the bisacodyl and something else happens in his bowel. What do you think the final magnesium, his reflexes and his urine will look like?

6T3 SBAR call. SBAR call to the night hospitalist about rebound magnesium and constipation.

	Say it out loud, then write it (short phrases and numbers)
S	
B	
A	
R	

CLUE UNLOCKED #6

The **accomplice** was a standing order. Teams that catch hidden magnesium in a protocol prevent a second crisis. **Question any order or protocol that contains magnesium in a kidney patient.**

POINTS ON THIS PAGE

- Clinical reasoning: ___ / 1 (of 6)
- Safety, priorities, SBAR: ___ / 1 (of 5)
- Teamwork, Complete: whole packet

STOP

Wait for the instructor's last 4h

Turn in with a paperclip (no staples). Write in dark pen. Do not turn the page until the instructor advances the clock.

CASE CLOSED: THE FINAL CHECK

Wednesday 0700

The clock has reached the end of the case. Check Ray one more time, then close the file.

SITUATION (the instructor reads this aloud)

It is **Wednesday 0700**. After the bisacodyl suppository Ray had a large stool at 0345 and slept. The 0500 magnesium is **5.3** (down from 6.4) and the creatinine is 3.3 (it peaked at 5.3). Urine has recovered to 85 mL/h. Weight is 90.0 kg (88.0 on arrival). Dr. Raman says no more dialysis for now. The dialysis catheter stays 24 hours just in case. The Foley can come out, and furosemide is due at 0800 for the extra fluid. Eleanor is back at the bedside.

C BEDSIDE NURSE

Initials: _____

Final check at the bedside (everyone looks, the Bedside Nurse writes).

Time	Temp (F)	HR	BP (MAP)	RR	SpO2 / oxygen	Pain	Rhythm (read the monitor)	Glucose	Weight
last time 0300	98.0	56	104/60 (75)	12	95% RA	4/10	Sinus rhythm 56, PR 0.24 s (long again)	126	-
0700 NOW									

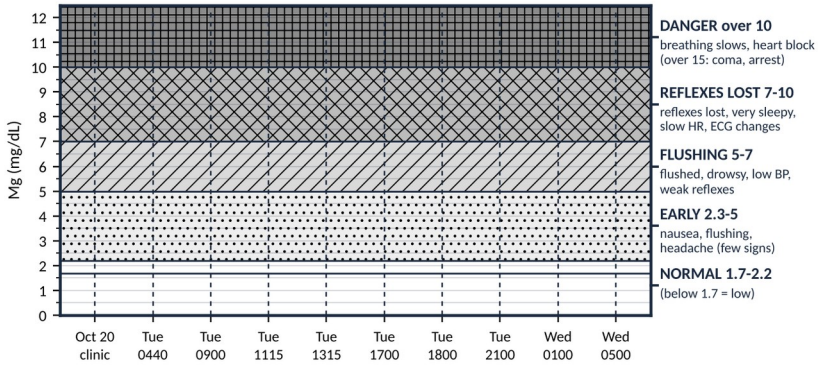
What to check	What you find	Better than 0300?
Level of consciousness and orientation		☐ yes ☐ no
Breathing: rate for a FULL minute		☐ yes ☐ no
Reflexes (0 to 4+) and grip		☐ yes ☐ no
Lungs, neck veins, ankles		☐ yes ☐ no
Abdomen and last stool		☐ yes ☐ no
Urine last hour; weight (kg)		☐ yes ☐ no

B Lab Interpreter: the last results, peak versus now.

Test	Normal	WORST value (circle when it happened)	Wed 0500 (final)
Magnesium (mg/dL)	1.7-2.2		
Potassium (mmol/L)	3.5-5.1		
Creatinine (mg/dL)	0.7-1.3		

MG-O-METER, THE WHOLE STORY: plot all your Mg results and connect the dots

from your Mg tracker boxes



Mark an X at each level, then write its zone below.

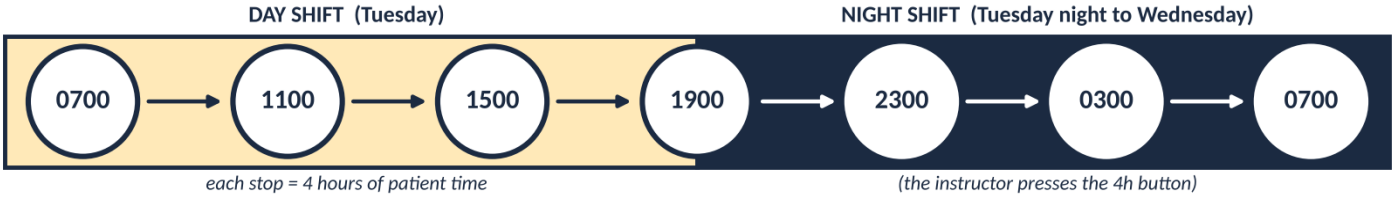
Mark an arrow on the chart for each of these: **calcium gluconate** (first given), **dialysis**, and the **hidden magnesium** you caught.

Turn in with a paperclip (no staples). Write in dark pen. Do not turn the page until the instructor advances the clock.

THE STAY IN SEVEN STOPS

Fill in the Mg result for each stop (use the lab draw closest to the time), then the big problem and what your squad did about it.

timeline



Stop	Mg result	The BIG problem at that stop	What our squad did
Tue 0700			
Tue 1100			
Tue 1500			
Tue 1900			
Tue 2300			
Wed 0300			
Wed 0700			

Plot-twist scoreboard. Look back at your six predictions. Which one came closest to the truth? What surprised you most?

Our top clue: which of the six "CLUE UNLOCKED" cards mattered most for keeping Ray safe, and why?

Turn in with a paperclip (no staples). Write in dark pen. Do not turn the page until the instructor advances the clock.

TEACH-BACK, REFLECTION AND TEAM CHECK

finish together, then individually

TEACH-BACK TO ELEANOR

Eleanor says: **"What do I do with all those bottles at home? He gets so constipated from his back medicine, and the heartburn medicine always helped."**

Write **three tips in plain words** (like you would say them at the kitchen table: no medical jargon). Think: which products to avoid and how she will know, safe choices, and when to call.

Tip 1	
Tip 2	
Tip 3	

How will you know Eleanor understood? Write your teach-back question (a question that makes her show you, not just say "yes"):

INDIVIDUAL REFLECTION (cut or fold along the dotted lines; write your own name)

Name: _____ **I did well:** _____

One thing I will do differently on clinical: _____

One question I still have: _____

Name: _____ **I did well:** _____

One thing I will do differently on clinical: _____

One question I still have: _____

Name: _____ **I did well:** _____

One thing I will do differently on clinical: _____

One question I still have: _____

PEER TEAMWORK CHECK (everyone fills in their own; the instructor sees it, your teammates do not)

Teammate (not you)	Did their share of the work	Listened to teammates	Spoke up with a good clue or question	One kind, useful comment
	☐ yes ☐ sometimes	☐ yes ☐ sometimes	☐ yes ☐ sometimes	
	☐ yes ☐ sometimes	☐ yes ☐ sometimes	☐ yes ☐ sometimes	

Rater name: _____